

YINGAO (ELAINE) YAO

elaine@cs.cornell.edu

<https://elaineyao.github.io/>

RESEARCH INTERESTS

My research interests are at the intersection of Software Engineering and Machine Learning (ML). I am particularly interested in developing software testing techniques (e.g., fuzzing) to improve the security, reliability, and correctness of ML systems.

EDUCATION

Cornell University *September 2024 - Present*

Ph.D. in Computer Science.

Advisor: Dr. Saikat Dutta.

University of British Columbia *September 2021 - December 2023*

MASc. Electrical and Computer Engineering.

Advisor: Dr. Karthik Pattabiraman.

University of Electronic Science and Technology of China *September 2017 - June 2021*

B.E. Electrical and Communication Engineering

GPA: 3.96/4.0

PUBLICATIONS

Automated Assertion Generation and Regression Testing for Machine Learning Notebooks *ASE 2026*

Yingao (Elaine) Yao, Vedant Nimje, Varun Viswanath, and Saikat Dutta.

IEEE/ACM International Conference on Automated Software Engineering.

SwarmFuzz: Discovering GPS Spoofing Attacks in Drone Swarms *DSN 2023*

Yingao (Elaine) Yao, Pritam Dash, and Karthik Pattabiraman.

IEEE/IFIP International Conference on Dependable Systems and Networks (DSN).

Poster: May the Swarm Be With You: Sensor Spoofing Attacks Against Drone Swarms *CCS 2023*

Yingao (Elaine) Yao, Pritam Dash, and Karthik Pattabiraman.

ACM SIGSAC Conference on Computer and Communications Security (CCS).

PREPRINTS

(Under submission) ICS-Sniper: A Targeted Blackhole Attack on Encrypted ICS Traffic *arXiv 2023*

Gargi Mitra, Pritam Dash, **Yingao Elaine Yao**, Aastha Mehta, and Karthik Pattabiraman.

<https://arxiv.org/abs/2312.06140>.

HONORS AND AWARDS

IEEE DSN Student Travel Grant Award *2023*

Faculty of Applied Science Graduate Award, University of British Columbia *2022*

Outstanding Graduate, Sichuan Province *2021*

Outstanding Winner, COMAP Interdisciplinary Contest in Modeling *2020*

Thanksgiving Scholarship for Modern Scientists *2020*

National Scholarship *2020*

National Second Prize in China Undergraduate Mathematical Contest in Modeling *2018*

WORKING EXPERIENCE

Security Researcher
Fortinet, Vancouver, Canada.

Jan 2024 - Aug 2024

TEACHING EXPERIENCE

Teaching Assistant for CS 5150: Software Engineering
Cornell University, Ithaca, NY, USA

Spring 2025

Teaching Assistant for CPEN 321: Software Engineering *Spring 2022, Fall 2022, Spring 2023*
University of British Columbia, Vancouver, Canada.

ACADEMIC SERVICE

Artifact Evaluation Program Committee

2024

19th USENIX Symposium on Operating Systems Design and Implementation (OSDI).

European Conference on Object-Oriented Programming (ECOOP).

18th USENIX WOOT Conference on Offensive Technologies (WOOT).

ACM Conference on Computer and Communications Security (CCS).

USENIX Annual Technical Conference (USENIX ATC).

The 54th Annual IEEE/IFIP International Conference on Dependable Systems and Networks (DSN).

Artifact Evaluation Program Committee

2023

The 34th IEEE International Symposium on Software Reliability Engineering (ISSRE).

DEPARTMENT SERVICE

Department of Computer Science, Picnic Czar
Cornell University, Ithaca, NY.

Fall 2025, Spring 2026

MENTORING

David Jovic (BS, University of Belgrade)

Arpan Chakraborty (BS, Adamas University)

Vedant Ramesh Nimje (BS, Veermata Jijabai Technological Institute, Mumbai)

Varun Viswanath (BS, Dwarkadas J Sanghvi College of Engineering)

Parshwa Nandu (BS, Dwarkadas J Sanghvi College of Engineering)

PRESENTATIONS

SwarmFuzz: Discovering GPS Spoofing Attacks in Drone Swarms

DSN 2023

Poster: May the Swarm Be With You: Sensor Spoofing Attacks Against Drone Swarms.

CCS 2023

OPEN-SOURCE CONTRIBUTIONS

SwarmFuzz: <https://github.com/DependableSystemsLab/SwarmFuzz>

NBTest: <https://github.com/seal-research/NBTest>